



Targeting LKB1–AMPK–SIRT1–induced autophagy and mitophagy pathways improves cerebrovascular homeostasis in APP/PS1 mice

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ABSTRACT

Background: Alzheimer's disease (AD) is the most common and severe degenerative disorder of the central nervous system in the elderly, profoundly impacting patients' quality of life. However, effective therapeutic agents for AD are still lacking. Bazi Bushen capsule (BZBS) is a traditional Chinese herbal compound with potential neuroprotective effects, yet its underlying mechanisms remain poorly understood.

Methods: In this study, we utilized APP/PS1 transgenic mice to assess the therapeutic efficacy of BZBS. Initially, we evaluated the spatial learning and memory of the mice using the Barnes maze. The brain microcirculation was assessed through a small-animal ultrasound system, two-photon *in vivo* imaging, and micro-computed tomography angiography. Molecular, biochemical, and pathological analyses were conducted on brain tissues. Through network pharmacology, we identified potential intervention pathways and targets for BZBS in the treatment of AD, which we subsequently validated both *in vivo* and *in vitro*. Additionally, we employed molecular virtual docking screening and biolayer interferometry to elucidate the direct interactions of ginsenoside Rg5 and ginsenoside Ro in BZBS with AMPK and LKB1 proteins.

Results: The BZBS intervention significantly enhanced spatial learning and memory in APP/PS1 mice while decreasing A β deposition. Furthermore, BZBS protected cerebrovascular homeostasis and mitigated neuro-inflammation, as evidenced by decreased blood–brain barrier permeability, increased expression of tight-junction proteins, and restored cerebral blood flow. Mechanistically, ginsenosides Rg5 and Ro in BZBS directly bind to AMPK and LKB1 proteins, activating the LKB1–AMPK–SIRT1 signaling pathway, promoting autophagy and mitochondrial autophagy, and alleviating oxidative stress damage in endothelial cells.

Conclusions: BZBS enhances autophagy-related activity, decreases A β deposition, and improves endothelial cell homeostasis through the activation of the LKB1–AMPK–SIRT1 signaling pathway, ultimately leading to improved cognitive function in mice with AD. This study highlights the importance of enhancing autophagic activity and maintaining cerebrovascular homeostasis in mitigating cognitive decline in AD, providing evidence and new insights into the application of compound medicines for treating age-related neurological disorders.

1. Introduction

Alzheimer's disease (AD), the most prevalent neurodegenerative

disorder [1], is characterized by a progressive decline in cognitive functions. By 2050, the global prevalence of AD is projected to triple, resulting in 152 million affected individuals and posing significant

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health and economic burdens [2]. Currently, the primary strategy for preventing and treating AD involves inhibiting the production and enhancing the clearance of A β [3]. However, the pathogenesis of AD is complex. There is emerging evidence suggesting that disruptions in cerebral vessels homeostasis play a crucial role in the onset and progression of AD [4]. Cerebral vessels are responsible for transporting oxygen and removing waste products from the brain. Importantly, cerebral vessels possess the blood–brain barrier (BBB)—a dynamic structure comprising brain microvascular endothelial cells, pericytes, astrocyte foot processes, basement membrane, and tight junctions—which prevents harmful substances in the blood from entering the brain [5]. Detrimental factors such as extravascular deposition of A β and oxidative stress can increase the permeability of BBB, leading to albumin leakage and neuroinflammation [6,7]. For example, stimulation by albumin activates microglia, which exhibits a proinflammatory phenotype and release inflammatory cytokines such as tumor necrosis factor α (TNF- α) and interleukin-1 β (IL-1 β). Simultaneously, microglia activate matrix metalloproteinases (MMPs), leading to additional damage to the BBB and impairing brain function [8,9]. It has been shown that the disruption of the BBB precedes the deposition of A β plaques and the emergence of cognitive impairment-related clinical symptoms, serving as an early pathological hallmark of AD [10]. These trials underscore the significance of cerebral circulation in the pathogenesis of AD.

Autophagy represents another promising therapeutic target for AD [11]. It is a tightly regulated intracellular catabolic process that clears damaged organelles (such as mitochondria [mitophagy]) and degrades abnormal protein aggregates [12,13]. Defects in autophagy frequently manifest in the early stages of AD, potentially contributing to the accelerated accumulation of abnormal protein aggregates [14]. Autophagy depends on mechanisms coordinated by several key energy sensors. AMP-activated protein kinase (AMPK) is a critical metabolic regulator that preserves cellular energy balance. LKB1 acts as an upstream regulator of the AMPK signaling pathway, phosphorylating AMPK to activate its function [7,15,16]. Once activated, AMPK upregulates essential proteins in the autophagy pathway, including microtubule-associated protein 1 light chain 3 (LC3B) and Beclin-1 [17,18]. Prior studies have confirmed that autophagy enhancers can accelerate A β degradation and ameliorate cognitive deficits *in vivo* [7,19]. However, further research is necessary to investigate the types of autophagy occurring in various brain regions and cell types, along with their therapeutic potential.

Compound medicines exhibit a range of pharmacological effects, rendering them suitable for treating chronic diseases, especially age-related neurological disorders. Bazi Bushen capsule (BZBS) contains 14 herbal ingredients such as ginseng, cistanche, and rehmannia. It is a proprietary medicine approved by the China National Medical Products Administration and demonstrates potential anti-aging properties in motor, metabolism, and digestive systems [20–27]. A previous study demonstrated that BZBS regulates DNA methylation status and rejuvenates the DNA methylation age in mice [28]. Clinical studies have indicated that BZBS alleviates symptoms of premature aging in humans, as demonstrated by improvements in grip strength, walking speed, and muscle mass [29]. More importantly, BZBS has been shown to improve cognitive dysfunction associated with aging [30]. However, the potential of BZBS to treat AD has not been studied. The aim of this study was to investigate the effectiveness of BZBS in improving cognitive impairment in mouse models of AD and its underlying mechanisms, thereby providing new insights into the use of combination drugs to treat neurodegenerative diseases.

2. Materials and methods

2.1. Animal models and drugs

APP/PS1 (APP^{swe}/PSEN1^{dE9}) double transgenic mice (3 months old, 24–27 g, female) were purchased from Zishen Technology Co., Ltd.,

and housed at the Hebei Yiling Traditional Chinese Medicine Research Institute under conditions of 25 °C, a 12-h dark–light cycle, and free access to water and food. The mouse study protocol was approved by the Ethics Committee of Hebei Yiling Traditional Chinese Medicine Research Institute (N2021174). Donepezil improves AD by inhibiting acetylcholinesterase (AChE). It was approved by the Food and Drug Administration (FDA) for the treatment of AD in 1996; therefore, we used Donepezil as the positive control drug [31,32]. Fifty mice with similar body weights were randomly divided into five groups, namely the wild type (WT), APP/PS1 (model), APP/PS1 + 1 mg/kg BZBS (BZ-low), APP/PS1 + 2 mg/kg BZBS (BZ-high), and APP/PS1 + 5 mg/kg donepezil (donepezil) groups. BZBS (Lot: B2006011, Shijiazhuang, China) and donepezil (Lot: H20070181, Suzhou, China) were suspended in 0.5 % sodium carboxymethylcellulose (CMC-Na) separately. The dosage, duration and method of administration of BZBS and donepezil was determined based on previous studies [20,28,33]. The mice from the WT and APP/PS1 groups were intragastrically administered with an equal volume of 0.5 % CMC-Na.

2.2. Barnes maze test

The Barnes maze (JLBehv-BNSG, Shanghai JiLiang, Shanghai, China) was utilized to evaluate spatial memory and ability. The apparatus features a circular maze that is 80 cm wide and contains 20 holes. Prior to the formal test, the mice underwent training sessions twice daily for three consecutive days. Following the initiation of training, the mice were confined in a light-tight box positioned at the center of the maze for 5 s to restrict their movement. Upon removal of the light-tight box, the timer commenced, enabling the mice to freely explore the maze for 180 s until they entered the escape hole. Once all limbs entered the target box, the mouse remained there for 30 s. If the mouse failed to locate the target box, it was removed from the maze and placed inside the target box for 30 s. Each trial was recorded using a camera positioned above the platform. During the official test, the mouse was confined in the light-tight box for 5 s; subsequent to this, the box was removed, and the recorded video was analyzed using DigBehv software (JLBehv-BNSG, Shanghai Kira, Shanghai, China).

2.3. Nissl staining

Nissl staining was performed in accordance with the manufacturer's instructions using the Nissl staining kit (Beyotime, Shanghai, China). In short, the sections were placed in toluidine blue, placed at 60 °C for 40 min, and then washed with distilled water. After rapid differentiation in 95 % ethanol, the sections were transferred to anhydrous ethanol for rapid dehydration. The results were observed under an optical microscope.

2.4. Two-photon microscopy

To measure BBB permeability, the mice were anesthetized and placed in a head holder under a custom-modified two-photon microscope (Scientifica, Uckfield, United Kingdom). A mixture of dyes containing 12.5 mg/mL fluorescein isothiocyanate (FITC)-conjugated dextran (70-kDa molecular weight, Sigma-Aldrich, United States) and 6.25 mg/mL tetramethylrhodamine (TRITC)-conjugated dextran (40 kDa, ThermoFisher Scientific, United States) was administered via tail vein injection. *In vivo* images were then acquired. Both 40 kDa and 70 kDa dyes were utilized to label blood vessels. When blood vessels in the brain become leaky, smaller molecular weight dyes tend to escape the vessels, whereas larger molecular weight dyes are retained within them [34].

2.5. Brain blood flow dynamics measurement

The mice were anesthetized with 2.5 % isoflurane in an induction

box and then placed in a supine position on a physiological information monitoring platform, with 1 % isoflurane to maintain anesthesia. The Vevo®3100LT ultrasound imaging platform was used to analyze blood flow dynamics. The middle cerebral artery and the posterior cerebral artery were measured three times each, and the average was taken. If a particular artery was too small to obtain clear blood flow and could not be repeatedly collected (more than three times per day, 3 days), then the MeanVel of that artery was considered 0. The average of the three arterial blood flow dynamics indicators was calculated for each animal.

2.6. Brain micro-computed tomography (micro-CT)

The mice were anesthetized with sodium pentobarbital and injected with 4 % paraformaldehyde. The prepared MICROFIL® vascular contrast agent was perfused at a rate of 0.3–0.5 mL/min. After hardening, the brain samples were removed, placed in 10 % phosphate buffer, and decalcified with 8 % formic acid for 48 h. Micro-CT imaging was used for cerebral angiography. The parameters of the micro-CT system were as follows: filter, 0.5 mm aluminum; voltage, 50 kV; current, 100 μ A; field of view, 18 mm; and 360° scanning time, 57 min. After scanning, image reconstruction was performed using the system software, and the Caliper MicroCT Analysis Tools module in the AnalyzeDirect workstation was used to perform three-dimensional reconstruction of the cerebral blood vessels and measure the vascular volume.

2.7. Magnetic luminex assay

Serum levels of IL-1 α , IL-1 β , IL-6, TNF- α , MMP2, and MMP9 in the mice were measured using MagpixLuminex. The brain tissue of the mice was lysed with non-denaturing tissue/cell lysate (Solarbio, Beijing, China). The multifactor assay was performed with a mouse premixed multi-analyte kit (LXSAMSM-15, batch number: L137690, R&D, Minneapolis, USA; LXSAMSM-01, batch number: L137689, R&D, Minneapolis, USA), and the data were read within 90 min using a MagpixLuminex analyzer (Luminex, Austin, TX). The assay method followed the instructions for the reagents.

2.8. Indirect calorimetry

Indirect calorimetry was performed to assess energy expenditure. Metabolic trajectory profiling was performed using mice. Metabolic parameters were measured using the Phenomaster system (TSE Systems, Inc.). The metabolic cage environment was set at 25 °C, 55 % humidity, and a 12-h light–dark cycle. After approximately 24 h of acclimatization, 24 h of data were collected for subsequent analysis. During the measurement period, the mice had ad libitum access to food and water. The light–dark cycle was set at 12 h (lights on at 7:00 a.m.). The measurement interval was set every 18 min for the measurement of metabolic parameters such as oxygen consumption (VO₂), carbon dioxide production (VCO₂), respiratory exchange ratio (RER), and energy expenditure.

2.9. Immunofluorescence staining

Immunofluorescence staining was performed for A β _{1–40}, A β _{1–42}, PSD95, Iba-1, GFAP, LC3B, Albumin, TOM20, CD31, and PDGFR β . Samples were fixed in 4 % paraformaldehyde and then sectioned. The sections were antigen-retrieved with 0.01 M citrate buffer through microwave heating. After incubation with hydrogen peroxide solution, the sections were sealed with 5 % bovine serum albumin (BSA) at room temperature for 30 min. Then, the sections were incubated with the primary antibody (Table S1) at 4 °C overnight, and the secondary antibody (Table S1) was added for 40 min in the dark. DAPI (SouthernBiotech, 0100-20) was used for nuclear staining. High-resolution images were acquired using a confocal microscope (Zeiss, Germany, LSM710) and were analyzed with ImageJ.

In cell experiments, immunofluorescence staining was performed for LC3B, LKB1, AMPK, SIRT1 and PGC1 α . Cells were seeded in 35 mm confocal culture dishes, and then fixed with 4 % paraformaldehyde for 20 min, permeabilized with 0.2 % Triton X - 100 (T8200; Solebo Biotechnology Co., Ltd.) for 10 min, and blocked with 5 % BSA for 30 min. The subsequent experimental procedures are consistent with those for tissue sections.

2.10. Real-time quantitative polymerase chain reaction (RT-qPCR)

The expression of IL-1 α , IL-1 β , IL-6, TNF- α , and MMP2 genes was determined by quantitative RT-qPCR. Total RNA was extracted from the brain tissue using the TransZolUpPlusRNAKit (TransGenBiotech, ER501-01, China), and cDNA synthesis was performed using the GoScript™ Reverse Transcription System (Promega, A5001, United States). Then, quantitative real-time PCR was carried out using the MonAmp™ChemoHSqPCRMix kit (Monad, MQ00401S, China). All experiments followed the manufacturer's instructions. Primers are shown in Table S2. The standard procedure was the same as that used in our previous studies [20].

2.11. Western blot (WB)

The expressions of ZO-1, Occludin, P62, Beclin1, LC3I/II, p-LKB1, p-AMPK, LKB1, and AMPK were determined by Western blot analysis of the brain cortex. The proteins were extracted from the brain cortex with high-efficiency RIPA tissue/cell lysate and PMSF (Solarbio, Beijing, China). The protein concentrations were estimated with a BCA protein assay kit (Beyotime Biotechnology, Shanghai, China). In each well, 40 μ g of protein was loaded and separated by electrophoresis at 120 V in precast 4 %–20 % polyacrylamide gels (GenScript Biotech, Nanjing, China) and transferred to a nitrocellulose membrane (Millipore, Billerica, MA).

After 1 h at 37 °C in a blocking solution (Beyotime Biotechnology, Shanghai, China), primary antibodies (Table S3) were added to the membrane for overnight incubation at 4 °C. On the next day, secondary antibody (Table S3) was added to the membrane at 37 °C for 1 h away from light. The Odyssey imaging system and image analysis software (LI-COR Biosciences, Lincoln, NE) were used to quantify the amount of target protein by grey scanning, using β -actin and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) as an endogenous reference protein [20].

2.12. Preparation of BZBS extract

BZBS ultrafine powder was mixed with 50 % ethanol (ratio 1:15) in an ultrasonic instrument (SONICS, USA). Then, it was centrifuged at 3500 rpm for 10 min, and the supernatant was collected and freeze-dried in a freeze dryer (LaboGene, Denmark, CS110-4) to prepare BZBS extract. The maximum dissolution concentration of BZBS extract was 50 mg/mL.

2.13. Cell culture

The mouse brain endothelial cell line bEnd.3 was purchased from the American Type Culture Collection (ATCC, Manassas, VA, USA) and cultured in 4.5 g/L glucose DMEM (Gibco, Invitrogen, USA) supplemented with 10 % Fetal Bovine Serum (FBS), 2 mmol/L glutamine, 100 units/mL penicillin, and 100 μ g/mL streptomycin after thawing. It was incubated at 37 °C and 5 % CO₂. The medium was changed every 2 days. The cells in the logarithmic growth phase were seeded on the plate at a density of 2×10^5 /mL and grown for 24 h before treatment. A β _{25–35} (HY-P0128) was purchased from Medchem Express, and the AMPK inhibitor (compound C dihydrochloride, CpC, T6146) and SIRT1 inhibitor (SIRT1-IN-1, T9648) were purchased from TargetMol. Ginsenoside Rg5 (B21044) and ginsenoside Ro (B21068) were purchased from Shanghai

Yuanze Biotechnology Co., Ltd. The bEnd.3 cells were divided into the normal group (control group), the model group (30 μM A β_{25-35}), the BZBS group (30 μM A β_{25-35} + 60 $\mu\text{g/mL}$ BZBS, 24 h) [35], the BZBS + CpC group (30 μM A β_{25-35} + 60 $\mu\text{g/mL}$ BZBS + 4 μM CpC, 24 h), the ginsenoside Rg5 group (30 μM A β_{25-35} + 20 μM ginsenoside Rg5, 24 h), the ginsenoside Ro group (30 μM A β_{25-35} + 20 μM ginsenoside Ro, 24 h), the ginsenoside Rg5 + CpC group (30 μM A β_{25-35} + 20 μM ginsenoside Rg5 + 4 μM CpC, 24 h), the ginsenoside Ro + CpC group (30 μM A β_{25-35} + 20 μM ginsenoside Ro + 4 μM CpC, 24 h), the ginsenoside Rg5 + SIRT1-IN-1 group (30 μM A β_{25-35} + 20 μM ginsenoside Rg5 + 1 nM SIRT1-IN-1, 24 h), and the ginsenoside Ro + SIRT1-IN-1 group (30 μM A β_{25-35} + 20 μM ginsenoside Ro + 1 nM SIRT1-IN-1, 24 h). The dosages selection of A β_{25-35} , BZBS, CpC, Mdivi-1, ginsenoside Rg5, ginsenoside Ro, and SIRT1-IN-1 was based on previous studies [35–40].

2.14. Cell viability assay

Cells were seeded into 96-well microplates at a density of 2×10^4 cells per well. Drug interventions were performed according to the above-described groups. Cell viability was determined using the Cell-Titer 96® Aqueous One Solution Cell Proliferation Assay Kit (G3581, Promega, USA). Each group in each independent experiment contained triplicate wells to ensure data reliability. The absorbance of the samples was measured at 450 nm.

2.15. Superoxide dismutase (SOD), Glutathione (GSH), and Malondialdehyde (MDA) assays

The plasma of the mice was used for SOD, MDA, and GSH assays. The levels of SOD, MDA, and GSH in the plasma samples were investigated using the Nanjing Jiancheng Bioengineering Institute SOD (A001-3-2), GSH (A005-1-2), and MDA (A003-1-2) assay kits. The absorbance at 450 nm, 412 nm, and 532 nm was detected by an ELISA reader (MR-96, Mindray, Shenzhen, China).

2.16. ATP measurement

The brain tissue of the mice was homogenized, and the supernatant was taken for determination. Using the ATP content determination kit (A095-1-1) of Nanjing Jiancheng Bioengineering Institute, the cell suspension was heated in a boiling water bath for 10 min, taken out and mixed and extracted for 1 min, and then used for determination. In accordance with the manufacturer's instructions, the level of ATP in the brain tissue and bEnd.3 cells was checked by the detection kit. The absorbance at 636 nm was detected by an ELISA reader (MR-96, Mindray, Shenzhen, China).

2.17. AMPK enzyme activity measurement

Cells were made into cell suspension for determination. The protein concentration was determined using the BCA protein detection kit (Beyotime Biotechnology, Shanghai, China). The cell AMPK enzyme activity spectrophotometric detection kit (Jiamei Biotechnology, Shanghai, China) was used to examine AMPK activity. The absorbance at 340 nm was measured at 0 and 5 min using a microplate reader (MR-96, Mindray, Shenzhen, China). A higher absorbance at 0 min than that at 5 min indicated enzyme activity.

2.18. Intracellular reactive oxygen species (ROS) measurement

Cells were stained using an ROS detection kit (Servicebio, G1706, China). After intervention on bEnd.3 cells according to the experimental protocol, the cells were washed with phosphate buffer saline (PBS). Then, the DCFH-DA probe solution (10 μM) was added, and the cells were incubated in the dark at 37 °C and 5 % CO₂ for 30 min. The cells were washed with PBS. Finally, images were acquired by confocal

scanning imaging microscopy (Zeiss, Germany, LSM710).

2.19. Mitochondrial membrane potential (MMP, $\Delta\Psi_m$)

Cells were washed with PBS, and the dye JC-1 (Servicebio, G1515, China) was used as a lipophilic cationic probe. Then, the JC-1 solution (1 μM) was added to each well and incubated for 10 min. Finally, images were captured with an inverted fluorescence microscope (Leica Microsystems, Wetzlar, Germany). The excitation wavelength was 485 nm, the emission wavelengths were 530 nm and 590 nm, and the fluorescence ratio of 590 nm (orange-red)/530 nm (green) was calculated.

2.20. Biolayer interferometry (BLI) binding assay

The binding affinities between ginsenosides Rg5, Ro, F3, and target proteins were evaluated with the BLI-based Octet Platform (Octet Red 96, ForteBio). An assay buffer consisted of 1 $\times$ PBS (pH = 7.4) containing 0.04 % Tween-20 and 0.1 % DMSO. Purified His-tagged LKB1 (Solarbio, P05989, 50 $\mu\text{g/mL}$) and human AMP Activated Alpha 2 (Feiyue Biotechnology, FY-P522491, 50 $\mu\text{g/mL}$) were individually immobilized on Ni-NTA biosensors (18–5101, ForteBio, State of California, USA) through three steps, namely equilibration, coupling, and again equilibration, leading to a saturation response of 4–6 nm. To determine the binding kinetics, the interactions between LKB1 and AMPK α 2 and the ginsenosides compounds (concentrations ranging from 3.125 to 100 μM , in twofold serial dilutions) were separately measured over 90 s for association and another 90 s for dissociation. Reference wells containing only buffer solution were included to correct the baseline shift. Another Ni-NTA control sensor was set up, which was incubated solely in the buffer, to provide negative reference controls to correct for any nonspecific binding of the compounds to the biosensor surface. A double reference subtraction approach, which involved subtracting both the background and nonspecific binding signals, was utilized to analyze the raw kinetic data. The binding affinity constant KD (where KD = kdis/kon, with kon representing the association rate constant and kdis representing the dissociation rate constant) values were determined with a 1:1 binding model through global fitting of multiple kinetic traces. GraphPad Prism (GraphPad Software, Inc., San Diego, USA) software was employed for the analysis of the real-time monitoring data.

2.21. Network pharmacology analysis

The comprehensive target spectrum of BZBS was obtained in accordance with the method described in previous research [35]. Using “Alzheimer's disease” and “cerebrovascular homeostasis” as keywords, the key genes were retrieved from six sources, including DisGeNET [41], Open Target Platform [42], MalaCards [43], CTD [44], GeneCards, and text mining [45]. DAVID 2021 (<https://david.ncifcrf.gov/>) was used for standardization of gene names [46]. To ensure the reliability of the data, only genes that appeared in more than three databases were retained as the core gene set for “Alzheimer's disease” or “cerebrovascular homeostasis.” The overlapping genes were analyzed to find the functional target of BZBS for treating cerebrovascular homeostasis in AD. To explain the mechanisms of BZBS against cerebrovascular homeostasis from a systematic perspective, we performed Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses using Metascape (<https://metascape.org>) and ClueGO plugin in Cytoscape [47].

2.22. Molecular docking

Molecular docking method was used to virtually evaluate the binding affinity between the compound and the target proteins, which might provide a reference for further experimental verification. AutoDock Vina 1.1.2 is an open-source program for molecular docking and virtual

screening [48]. SDF files of toxic molecules were downloaded from the PubChem database (<https://pubchem.ncbi.nlm.nih.gov/>) [49] and through the Open Babel 2.4.1 [50] into MOL2 format. The crystal structure of AMPK (PDB ID: 5EZV) and LKB1 (PDB ID: 5WXN) was obtained from the PDB database (<https://www.rcsb.org/>) [51]. We chose staurosporine (CID: 44259), the co-crystalline ligand of AMPK, as the positive control. Small molecules were energy-minimized by using PyRx-0.8 software. Ligands and receptors were prepared using the AutoDock Vina 1.1.2 tutorial. Using a rigid docking method, the grid properties during docking were determined based on the spatial position of the co-crystallized ligand binding in the protein. The parameters in the x, y, and z dimensions of AMPK's grid coordinates were set at 13.64, with a spacing of 0.375 Å. The positions of the x center, y center, and z center were 113.889, −81.054, and 33.485, respectively. The parameters in the x, y, and z dimensions of LKB1's grid coordinates were set at 12.0, 12.75, and 24.75, respectively, with a spacing of 0.375 Å. The positions of the x center, y center, and z center were 61.363, 75.275, and −7.046, respectively. The energy range was set at 3, and the exhaustivity was set at 8. For each structure, water molecules were removed, hydrogen atoms were added, Gasteiger charge was calculated, and the file was saved in the PDBQT format to complete ligand preparation. Generally, lower Vina scores indicated higher affinity between the ligand and the receptor. The general selection threshold was binding energy ≤ -5 kcal/mol [52,53]. Finally, PyMOL software was used to visualize the interaction mode between the compound and the target proteins.

2.23. Mitochondrial ROS (mito ROS) detection

The MitoSOX™ Red Mitochondrial Superoxide Indicators (M36006) from Invitrogen were used to detect mitochondrial ROS levels in bEnd.3 cells. The cells were incubated with 500 nM working solution at 37 °C for 30 min. Fluorescence was detected by a confocal microscope (Zeiss, Germany, LSM710) at 396 nm excitation and 501–610 nm emission. The average fluorescence intensity of mt ROS in bEnd.3 cells was analyzed using Image J software.

2.24. Statistical analysis

Data are represented as the mean $\pm$ standard deviation (SD), and all statistical analyses and graphing were performed using SPSS26.0 (SPSS, Armonk, United States) and GraphPad Prism8.0.2 (GraphPad, San Diego, United States). Data were checked for normal distribution (Shapiro–Wilk test) and homogeneity of variance (Levene's test for equality of variance) before choosing the appropriate statistical test. One-way ANOVA was used to determine differences between groups, and post hoc comparisons were made using the Tukey's post hoc test. A p value lower than 0.05 was considered significant.

3. Results

3.1. BZBS improves cognitive dysfunction and reduces amyloid deposition in APP/PS1 mice

We utilized the Barnes maze to evaluate spatial learning and memory capabilities across five groups. Compared with the model group, the BZBS-treated groups exhibited improved cognitive abilities, as evidenced by decreased latency (Fig. 1B), shorter total time (Fig. 1D), and reduced total distance traveled (Fig. 1C). Notably, BZBS demonstrates comparable efficacy to donepezil in enhancing memory in mice. Nissl bodies, which are plaque- or granule-like substances within neurons, serve as indicators of neuronal function [54]. Nissl staining revealed a reduction in Nissl body count in the hippocampal and cortical regions of the model group, whereas the BZBS-treated groups demonstrated an increase in the Nissl body number (Fig. 1E–G). Reduced synaptic density and plasticity directly contribute to cognitive decline [55]. PSD-95 is a

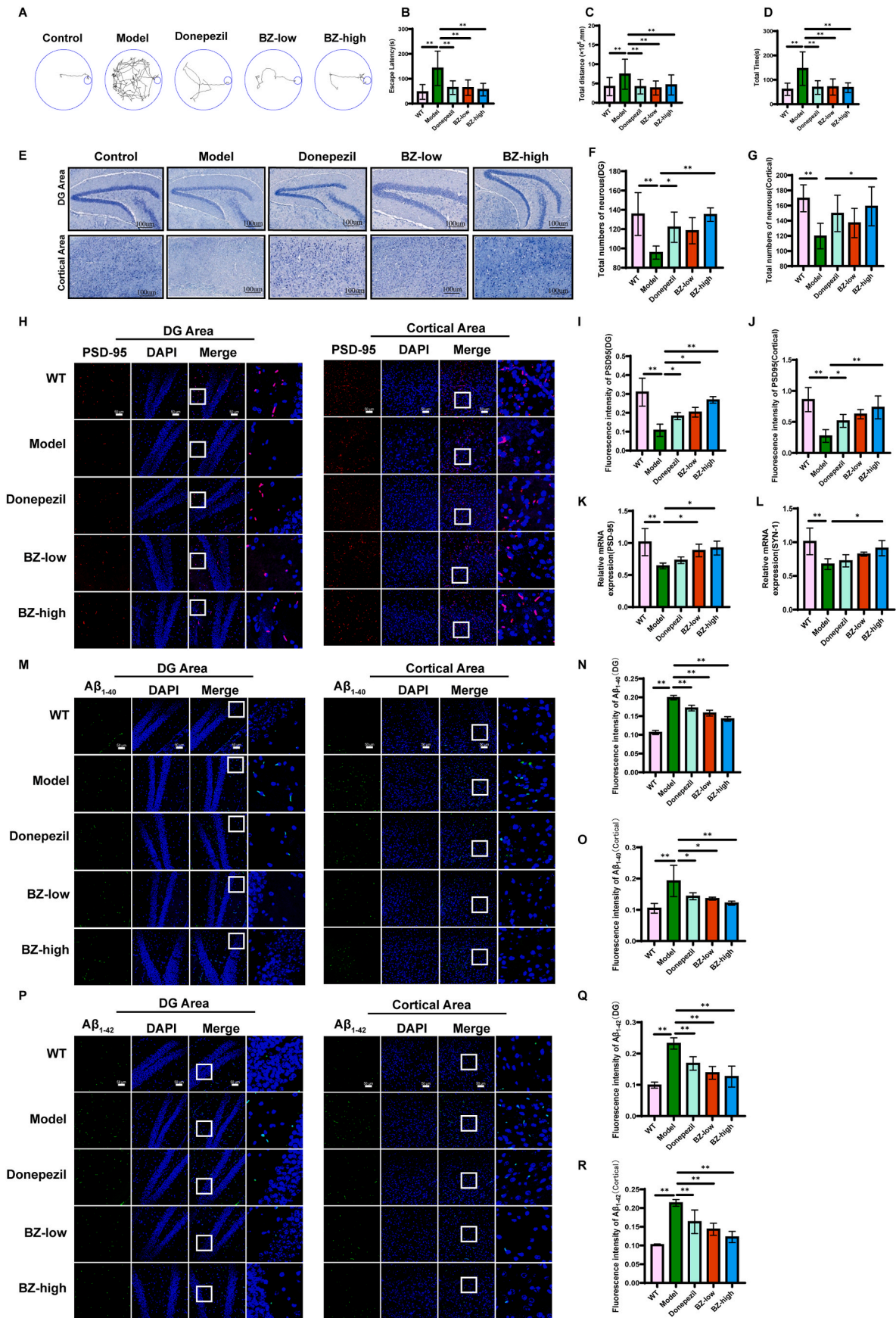
crucial protein required for maintaining the activity and stability of postsynaptic membrane receptors [56], and Synaptophysin (SYN), serving as a synaptic growth protein, is a well-established marker of synaptogenesis [57]. We detected a significant increase in immunofluorescence intensity of PSD-95 in the hippocampal and cortical regions of the BZBS-treated groups compared with the model group (Fig. 1H–J). The mRNA levels of PSD-95 and SYP were also significantly upregulated in the BZBS-treated groups compared with the model group (Fig. 1K and L). Furthermore, by immunofluorescence staining, we found that BZBS significantly decreased A β (both A β _{1–42} and A β _{1–40}, the main types of A β deposition around blood vessels) deposition in the hippocampus and cortex of APP/PS1 transgenic mice (Fig. 1M–R) [58,59]. Taken together, the results suggest that BZBS enhances cognitive function in AD mouse models.

3.2. BZBS improves the cerebrovascular structure and function in APP/PS1 mice

It has been shown in AD mouse models that A β (especially A β _{1–40})-induced damage to vascular endothelial cells disrupts the BBB, resulting in the leakage of blood components and worsening brain inflammation [58,59]. To investigate this hypothesis, we utilized *in vivo* two-photon microscopy to evaluate the function of the brain microvascular barrier in mice. The results demonstrated a significant leakage of 40 kDa TRITC-conjugated dextran in the brains of the model mice, while the BZBS group displayed a marked reduction in leakage (Fig. 2A and B). RT-qPCR and WB analyses revealed that, compared with the model group, the BZBS group showed significantly upregulated expression of tight-junction proteins ZO-1 and Occludin (Fig. 2C–G). These findings suggest that AD mice experienced disruption of the BBB, while BZBS effectively maintained cerebrovascular integrity. We further evaluated blood flow in the middle cerebral artery and the posterior cerebral artery of AD mice. Compared with the model group, the BZBS group showed a significant increase in end-diastolic blood flow velocity and pulsatility index in both the middle and posterior cerebral arteries (Fig. 2H–K). More importantly, the BZBS group demonstrated an increase in cerebral vascular volume and enhanced vessel integrity in AD mice, as assessed by micro-CT (Fig. 2L and M). These findings indicate that BZBS effectively restores the disrupted vascular homeostasis in the AD mouse model. Notably, BZBS exhibits a greater efficacy than the positive control drug, donepezil, in enhancing the BBB function and increasing cerebral blood flow velocity.

3.3. BZBS mitigates oxidative stress and neural inflammation in APP/PS1 mice

Dysfunction of cerebrovascular transport and the infiltration of toxic substances can result in oxidative stress and neuroinflammation in the brain [60–62]. Our analysis revealed a significant reduction in the levels of SOD and GSH in the brains of APP/PS1 mice, alongside elevated levels of MDA (Figs. S2A–C). BZBS demonstrated a dose-dependent decrease in oxidative stress within the brain. Furthermore, immunofluorescence staining of microglia and astrocytes in the hippocampal and cortical regions revealed a significant reduction in fluorescence intensity following BZBS treatment (Fig. 3A–H). Co-staining of albumin and microglia revealed that the model mice exhibited extensive albumin leakage and microglial phagocytosis, while these symptoms were mitigated in the BZBS-treated group (Fig. 3I–K). This is consistent with the previous studies of Giannoni P et al. [63]. We additionally measured the levels of inflammatory factors, particularly IL-1 α , IL-1 β , IL-6, TNF α , and MMP2, in the brain of the mice. We found that both mRNA and protein levels of proinflammatory factors were significantly elevated in the model group, indicating the presence of neuroinflammation in the AD mice. BZBS treatment significantly reduced the expression of proinflammatory cytokines (Fig. 3L–V). Our findings suggest that BZBS ameliorates oxidative stress and inflammatory responses in AD mice.



(caption on next page)

Fig. 1. BZBS alleviates cognitive impairment and A β deposition in AD mice. (A–D) Representative tracks, latency, total distance, and total time of the Barnes maze test demonstrated that BZBS markedly enhances learning and memory functions in AD mice ($n = 10$). (E–G) Analysis of Nissl body expression by Nissl staining revealed that BZBS significantly increased the number of Nissl bodies in the cortical and dentate gyrus (DG) regions of the hippocampus in AD mice. $n = 3$; scale bar = 100 μm . (H–J) Analysis of PSD-95 expression in the hippocampus and cortex by immunofluorescence showed that BZBS significantly elevated PSD-95 levels in both the cortical and DG regions of AD mice. $n = 3$; scale bar = 50 μm . (K–L) Relative mRNA expression of PSD-95 and SYN-1. $n = 3$ (M – O) Analysis of A β_{1-40} expression in the hippocampus DG region and cortex by immunofluorescence. $n = 3$; scale bar = 50 μm . (P–R) Analysis of A β_{1-40} expression in the hippocampus DG region and cortex by immunofluorescence confirmed that BZBS significantly reduced A β_{1-40} and A β_{1-42} expression in the DG and cortical regions of AD mice. $n = 3$; scale bar = 50 μm * $p < 0.05$, ** $p < 0.01$. Error bars represent SD.

3.4. BZBS enhances cerebrovascular autophagy and mitophagy in APP/PS1 mice

BZBS contains 14 herbs with a variety of bioactive compounds. To investigate the underlying mechanism of BZBS in the targeted therapy of AD, we integrated the components in the public herbal medicine database and obtained a total of 4080 unique compounds as the BZBS compound library. We also identified 4311 targets associated with BZBS compounds and 388 targets associated with AD (most of which are speculative targets) (Fig. 4A). Further analysis was performed on the common targets of BZBS and AD. The results revealed that the four most significantly enriched biological processes were energy acquisition through organic compound oxidation, ubiquitin-dependent proteolytic metabolism mediated by the proteasome, regulation of chemical synaptic transmission, and regulation of transsynaptic signaling. The four most significantly enriched cellular components were mitochondrial protein complexes, mitochondrial inner membranes, peptidase complexes, and endopeptidase complexes. The two most significantly enriched molecular functions were protein serine/threonine kinase activity and protein serine-specific kinase activity (Fig. 4B). KEGG enrichment analysis indicated that AD and neurodegenerative diseases were significantly enriched. Furthermore, the AMPK, TNF- α , neurotrophic factor, and NF- κB signaling pathways, as well as tight junctions, gap junctions, autophagy, apoptosis, cellular senescence, and oxidative phosphorylation, were also enriched (Fig. 4C).

Analyses in network pharmacology have emphasized the significance of autophagy in multiple target gene sets. Western blot analysis revealed that, compared with the model group, the BZBS group exhibited increased expression of Beclin-1 (a crucial regulator of autophagosome formation) and LC3II/LC3I (a marker indicative of autophagy activation). In contrast, the expression of SQSTM1/p62, a classic substrate of autophagy, was decreased in the BZBS group (Fig. 4D–G). This indicated a decrease in autophagy levels in the brain tissue of the model group, whereas BZBS treatment resulted in an increase in autophagy levels (Fig. 4D–G). Double immunofluorescence staining for LC3B and the endothelial cell marker CD31 or pericytes marker PDGFR β revealed that autophagy in the mice brain predominantly occurred near blood vessels and exhibited regional specificity, with a significant reduction in autophagy levels in the cortical area in the AD mice. Compared with the model group, BZBS treatment significantly elevated autophagy levels in the vicinity of blood vessels within the AD mouse brain (Fig. 4H–K). To assess whether BZBS influences cerebrovascular A β clearance in AD mice through autophagy, we conducted double immunofluorescence staining for LC3B and A β_{1-40} (the major form of A β deposits in cerebral microvessels and capillaries) across all mouse groups. Compared with the model group, the BZBS-treated AD mice exhibited elevated co-localization of LC3 and A β_{1-40} (Fig. 4L and M). Taken together, these results indicate that BZBS treatment enhances the autophagy levels of microvascular endothelial cells in the brains of AD mice.

Apart from autophagy, network pharmacology analysis also revealed a potentially significant impact of BZBS on mitochondrial function. We detected the ATP content in the brain tissue of the mice. Specifically, we found that the ATP content in the brain of the model group was significantly decreased, while the content increased after BZBS treatment, suggesting that BZBS improves mitochondrial function (Fig. S3A). Mitophagy, a process that involves the selective degradation of

mitochondria via autophagy, is crucial for maintaining mitochondrial homeostasis [64]. It typically occurs in defective mitochondria after damage or stress, and dysregulation of mitophagy is frequently observed in AD mice [65,66]. To assess mitochondrial autophagy in the brain vascular endothelium, double immunofluorescence staining for LC3B and TOM20 was performed in the hippocampus and cortex of each mouse group. The results indicated reduced co-localization between LC3B and TOM20 in the model group. BZBS treatment significantly enhanced the colocalization of these markers in the cortical region, but not in the hippocampus (Fig. 4N–O and Figs. S3B–H). Our findings suggest that BZBS enhances mitochondrial autophagy in the AD mouse brain.

3.5. BZBS regulates autophagy and mitophagy via activating the LKB1–AMPK–SIRT1 signaling pathway

Network pharmacology analysis indicated that BZBS functions primarily by modulating the activities of serine/threonine kinases and serine-specific kinases. WB analysis of mouse brains revealed that phosphorylation levels of AMPK and its upstream regulator LKB1 were significantly lower in the brains of AD mice than in those of the control-group mice. In contrast, BZBS treatment significantly elevated the phosphorylation levels of both LKB1 and AMPK (Fig. 5A–C). Indirect calorimetry to assess overall energy expenditure in mice also suggested that the basal metabolic rate increased after BZBS administration, suggesting potential AMPK activation (Figs. S4A–D). To confirm these findings, we treated bEnd.3 cells with BZBS (Fig. S5). Our results showed that incubation with BZBS increased the phosphorylation of LKB1 and AMPK, as well as LC3B positive vesicles (Fig. 5D–I). Additionally, BZBS treatment restored the A β_{25-35} -induced decline in mitochondrial membrane potential and ATP production while decreasing cellular ROS levels (Fig. 6A–E). Importantly, all BZBS-induced effects were reversed by co-incubation with AMPK inhibitors (Fig. 6A–G). And BZBS failed to reduce the mitochondrial ROS level together with LC3B inhibitor incubation (Figs. S6C–D). Following deacetylation of PGC1 α , SIRT1 translocates to the nucleus along with PGC1 α , thereby inducing the expression of mitophagy-related genes. Treatment with BZBS enhanced the nuclear localization of both SIRT1 and PGC1 α (Fig. 6F and G). In conclusion, our findings demonstrate that BZBS regulates cellular autophagy and mitophagy through the LKB1–AMPK–SIRT1 pathway, thereby maintaining homeostasis in vascular endothelial cells.

3.6. Identifying ginsenosides in BZBS as agonists of LKB1 and AMPK

To further explore the pharmacological mechanisms of BZBS, we conducted virtual docking screenings of the compounds targeting LKB1 and AMPK. The results revealed that 524 compounds exhibited significant binding affinity for AMPK, while 13 compounds targeted LKB1, defined by binding scores below -7 (Fig. 7A and B). Among these, ginsenosides Ro and F3 displayed a high binding affinity for both LKB1 and AMPK (Fig. 7C and D). Ginsenoside Rg5, a well-known ginsenoside involved in AMPK signaling regulation, exhibited binding scores of -6.9 for AMPK and -6.1 for LKB1 (Fig. 7A and B). Bio-layer interferometry analysis confirmed direct interactions of ginsenosides Ro and Rg5 with LKB1 and AMPK, whereas no interaction was observed for F3 (Fig. 7E–J). The binding affinities of ginsenosides for LKB1 were 51.59 μM (ginsenoside Rg5) and 46.19 μM (ginsenoside Ro), while for AMPK

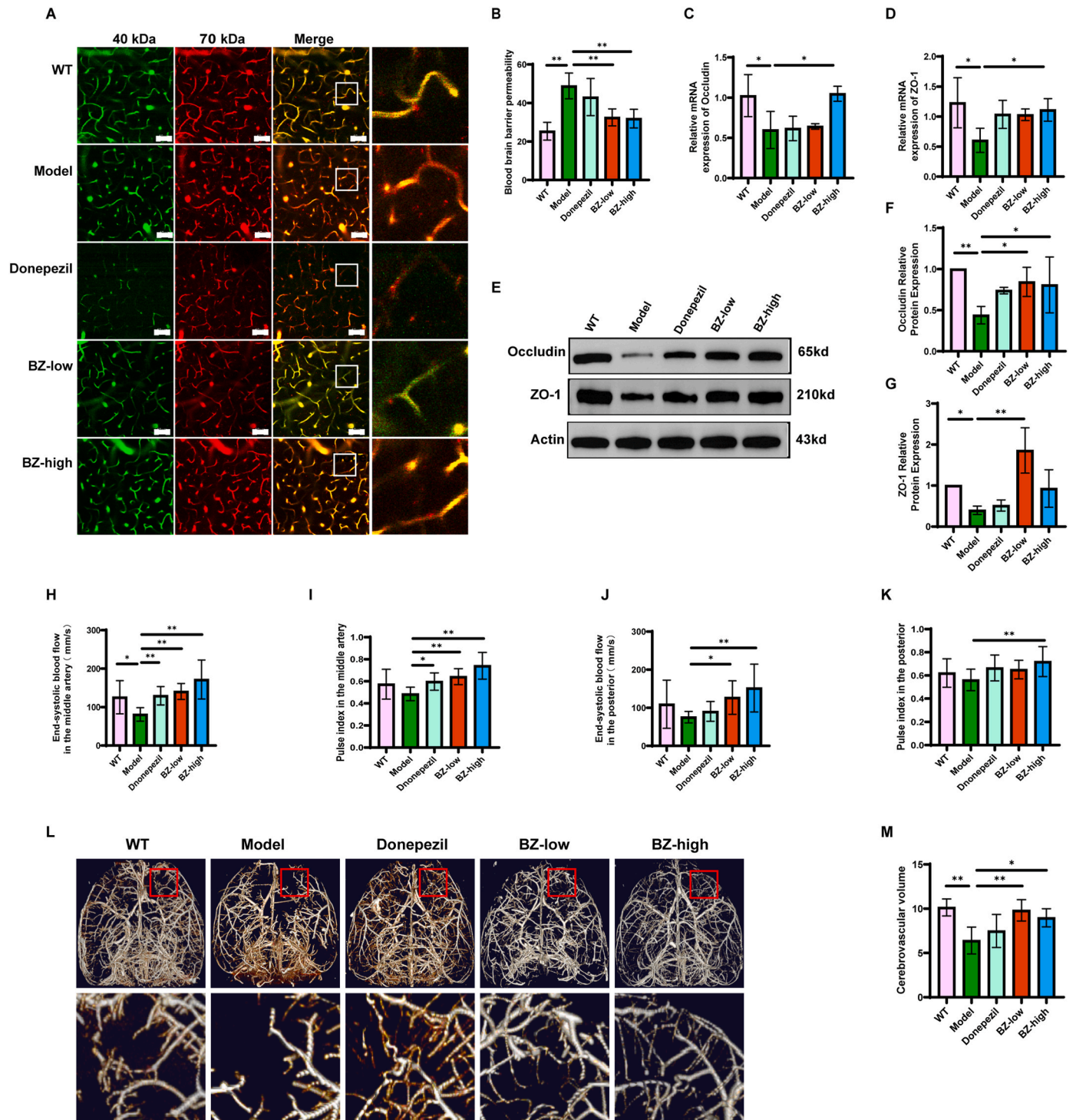


Fig. 2. BZBS improves BBB permeability and cerebrovascular density in AD mice. (A–B) The blood-brain barrier permeability was measured by detecting the exudation of 40 kDa small molecule dyes using two-photon imaging. BZBS significantly reduced the permeability of BBB. $n = 3$; scale bar = 50 μ m. (C–D) Relative mRNA expression of Occludin and ZO-1. $n = 3$. (E–G) Western blot analysis showed that BZBS significantly increased the expression of Occludin and ZO-1 proteins. $n = 3$. (H–K) Cerebral hemodynamics measuring end-systolic pressure and pulsatility index in the middle and posterior cerebral arteries. $n = 8$. (L–M) Cerebral vascular volume measured by micro-CT. $n = 3$; * $p < 0.05$, ** $p < 0.01$. Error bars represent SD.

they were 91.26 μ M (ginsenoside Rg5) and 11.78 μ M (ginsenoside Ro).

These findings suggest that ginsenosides in BZBS may enhance autophagy and mitophagy by activating LKB1 or AMPK. To test this hypothesis, we evaluated the safety and efficiency of ginsenosides in bEnd.3 cells (Figs. S7A–B). The results showed that ginsenoside Rg5 and ginsenoside Ro significantly increased the phosphorylation levels of LKB1 and AMPK (Fig. 8A–D). Consistent with the findings from BZBS

treatment, the effects of ginsenosides on the phosphorylation of LKB1 and AMPK were reversed by AMPK inhibitors (Fig. 8A–D). Furthermore, ginsenoside Rg5 and ginsenoside Ro enhanced AMPK activity *in vitro* (Fig. 8E). Treatment with ginsenosides Rg5 and Ro increased LC3B positive vesicles (Fig. 8F and G) and promoted the nuclear co-localization of SIRT1 and PGC1 α (Fig. 8H–I). Similarly, these effects were reversible with AMPK inhibitors (Fig. 8F–I). More importantly,

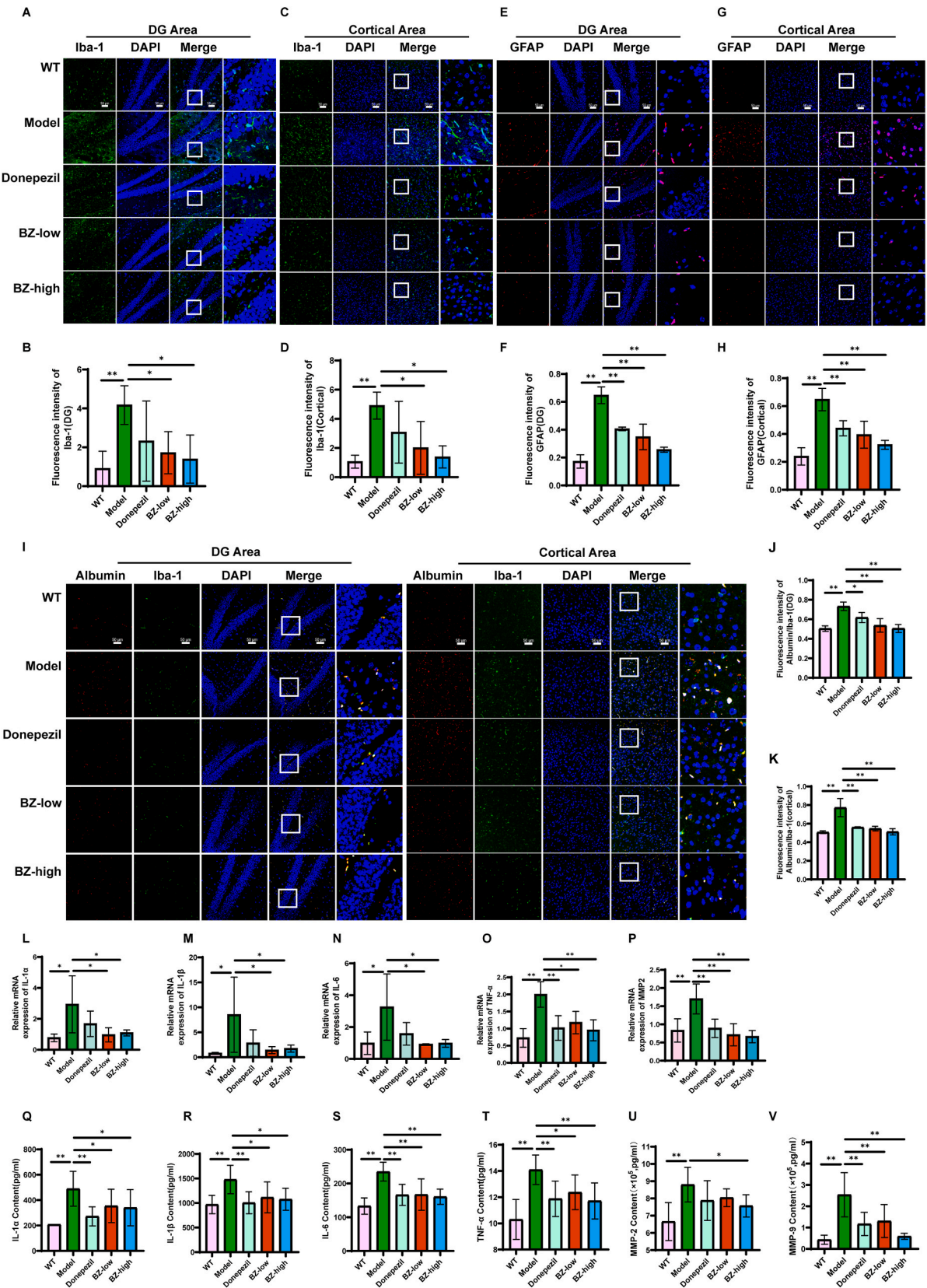
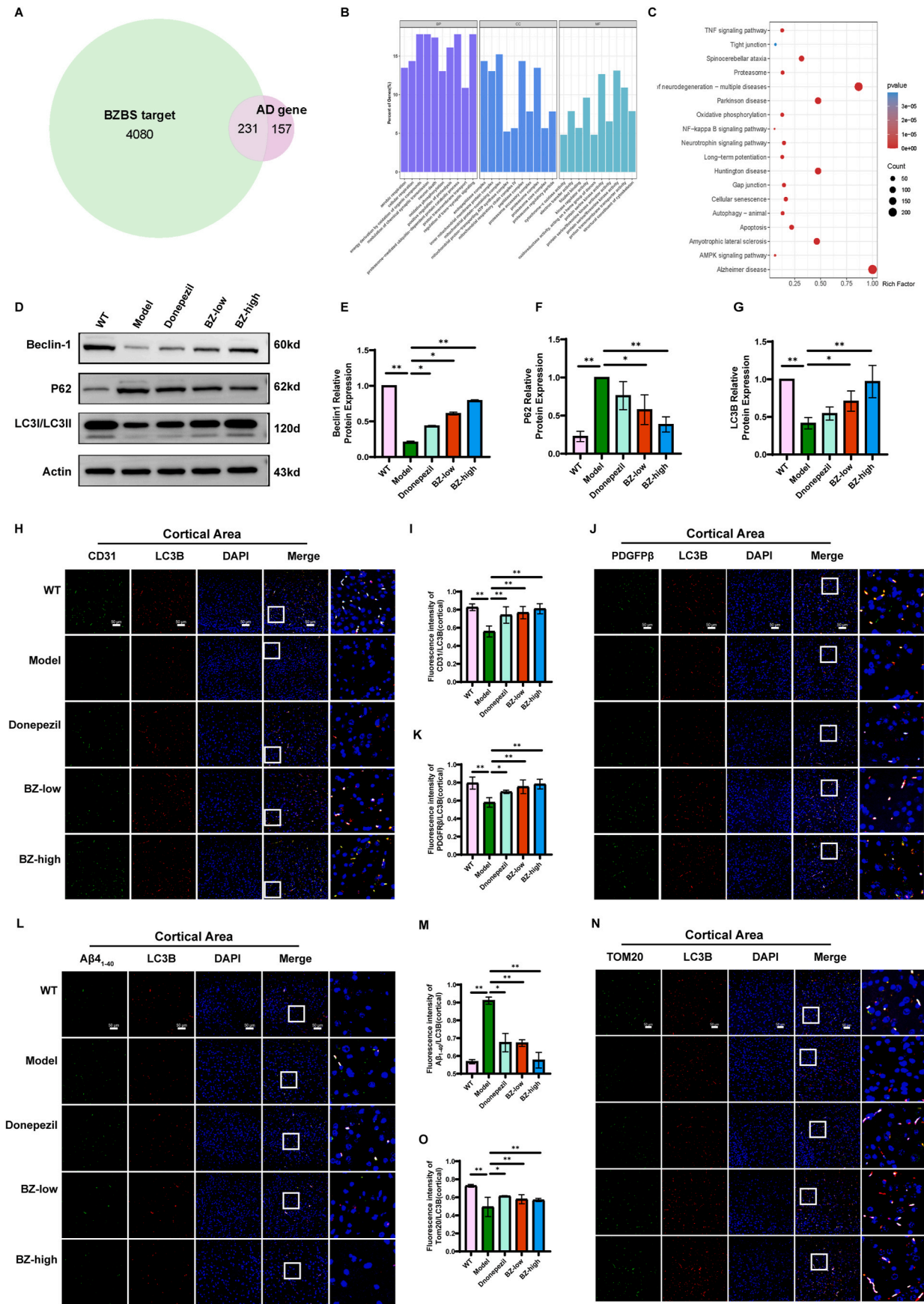


Fig. 3. BZBS reduces neuroinflammation in AD mice. (A–D) Expression of Iba-1 in the hippocampus DG region and cortex was analyzed by immunofluorescence. (E–H) Expression of GFAP in the hippocampus DG region and cortex was analyzed by immunofluorescence. (I–K) Expression of albumin and Iba-1 in the hippocampus and cortex was analyzed by immunofluorescence. $n = 3$; scale bar = 50 μm . (L–P) Relative mRNA expression of IL-1 α , IL-1 β , IL-6, TNF- α , and MMP-2. $n = 3$. (Q–V) Magnetic luminex assay was used to estimate the levels of IL-1 α , IL-1 β , IL-6, TNF- α , and MMP-2 in plasma. $n = 8$; * $p < 0.05$, ** $p < 0.01$. Error bars represent SD.



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Fig. 4. BZBS accelerates A β clearance by enhancing endothelial cell autophagy and mitochondrial autophagy. (A) Targets of the intersection of AD and BZBS. (B) GO enrichment analysis results. GO categories include “molecular function,” “cellular component,” and “biological process” terms. (C) KEGG pathway enrichment analysis results. The larger the rich factor, the higher the enrichment. The size of the dot is proportional to the number of genes enriched in the term. (D–G) The expression of Beclin-1, p62/SQSTM1, and LC3I/II proteins in the cerebral cortex of AD mice was detected by WB. BZBS significantly enhanced autophagy. $n = 3$. (H–I) Expression of CD31 and LC3B in the cortex was analyzed by immunofluorescence. (J–K) Expression of PDGFR β and LC3B in the cortex was analyzed by immunofluorescence. BZBS is likely to promote endothelial autophagy. (L–M) Expression of A β_{1-40} and LC3B in the cortex was analyzed by immunofluorescence. (N–O) Expression of TOM20 and LC3B in the cortex was analyzed by immunofluorescence. BZBS enhanced mitochondrial autophagy. $n = 3$; scale bar = 50 μ m; * $p < 0.05$, ** $p < 0.01$. Error bars represent SD.

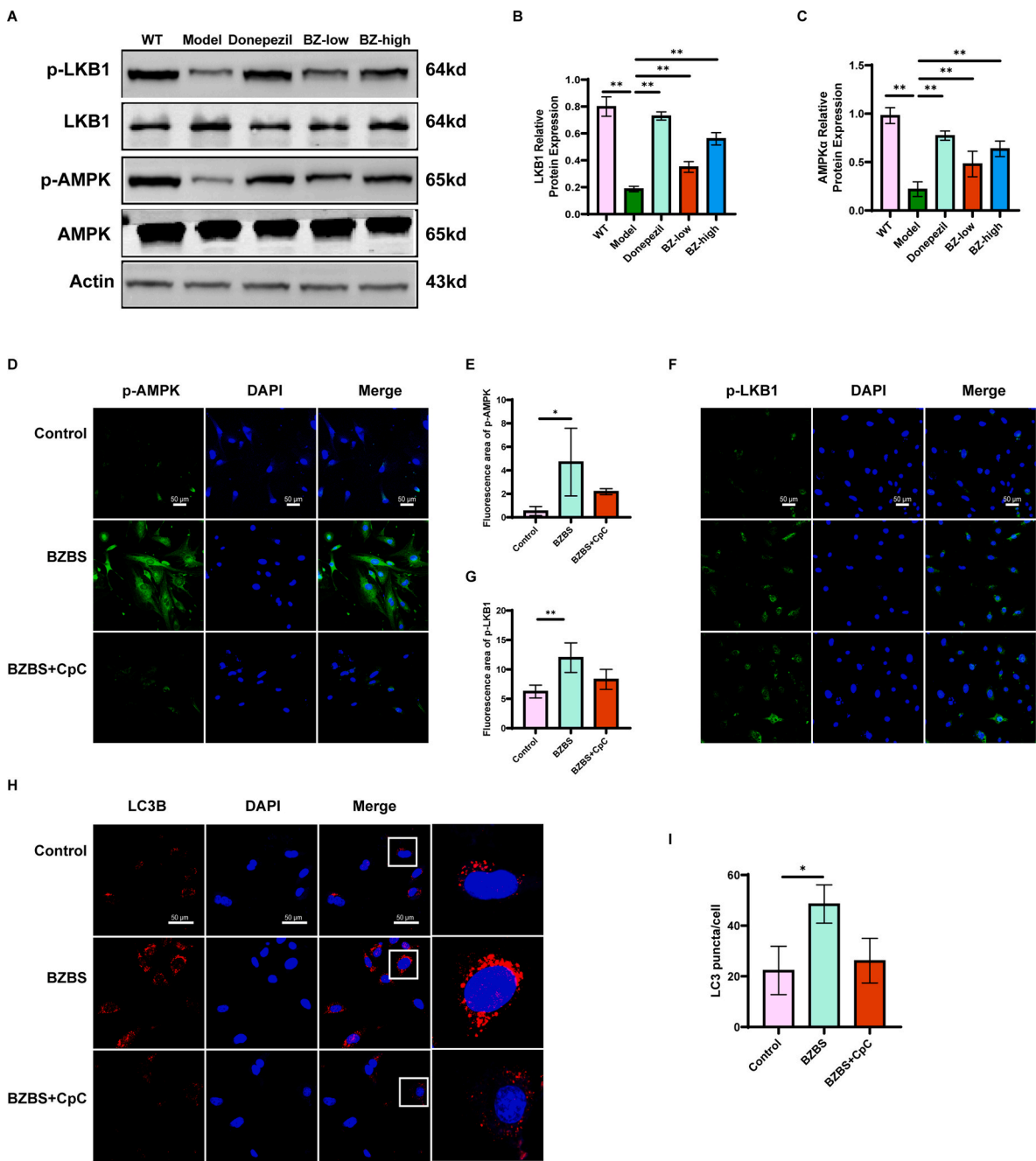


Fig. 5. BZBS enhances autophagy in bEnd.3 cells via the LKB1–AMPK–SIRT1 signaling pathway. (A–C) The expression of p-LKB1 and p-AMPK proteins in the cerebral cortex of AD mice was detected by WB. $n = 3$. (D–E) Expression of p-LKB1 in bEnd.3 cells analyzed by immunofluorescence. (F–G) Expression of p-AMPK in bEnd.3 cells analyzed by immunofluorescence. (H–I) LC3B Positive vesicles in bEnd.3 cells analyzed by immunofluorescence. $n = 3$; scale bar = 50 μ m; * $p < 0.05$, ** $p < 0.01$. Error bars represent SD.

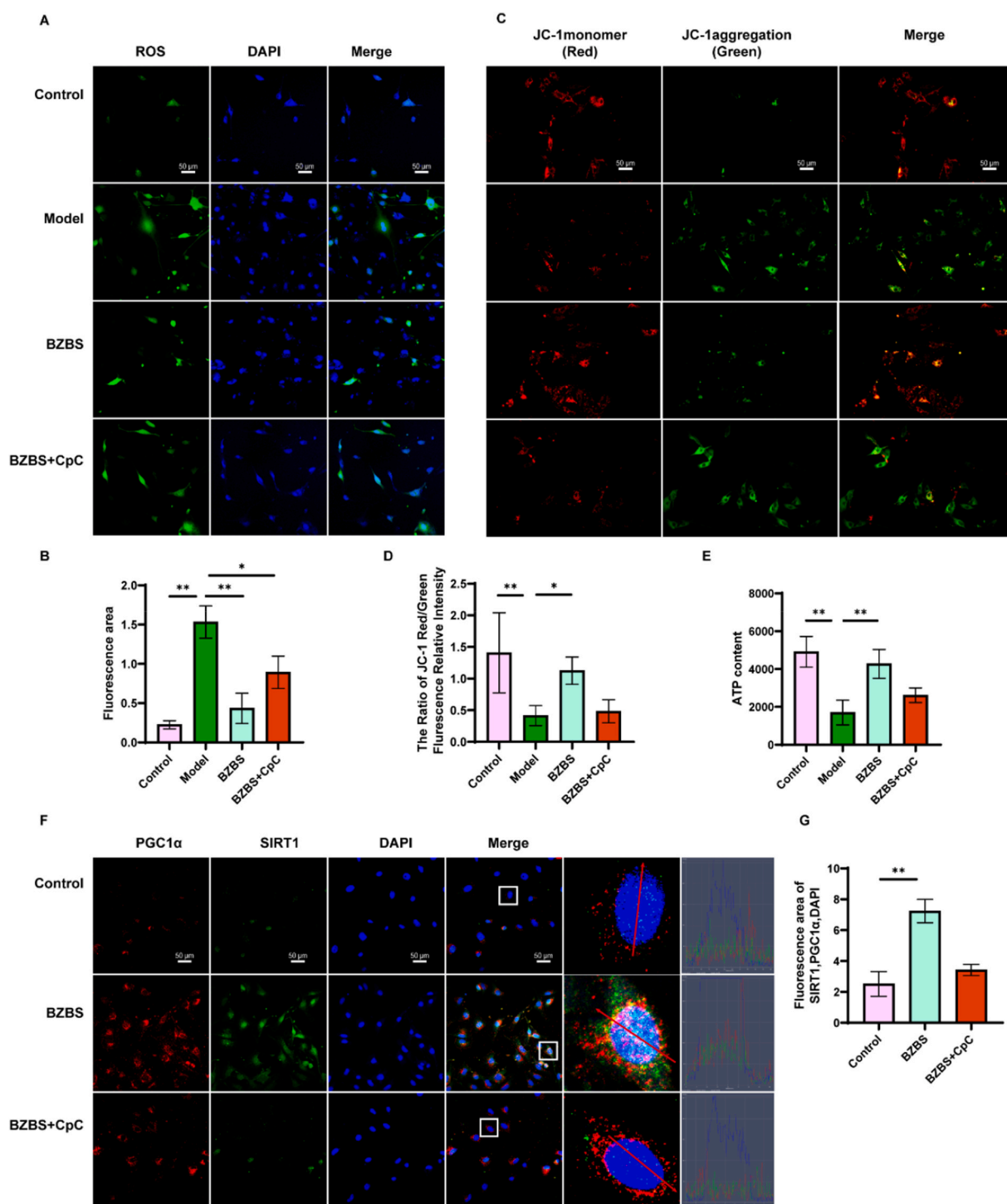


Fig. 6. BZBS alleviates mitochondrial dysfunction in bEnd.3 cells. (A–B) ROS measurement in bEnd.3 cells. (C–D) Mitochondrial membrane potential measurement in bEnd.3 cells. (E) ATP content measurement in bEnd.3 cells. (F–G) Co-expression of SIRT1, PGC1α, and DAPI in bEnd.3 cells by immunofluorescence analysis. $n = 3$; scale bar = 50 μm ; * $p < 0.05$, ** $p < 0.01$. Error bars represent SD.

ginsenosides Rg5 and Ro restored the A β_{25-35} -induced decline in mitochondrial membrane potential and ATP production (Fig. 9C–E) while decreasing cellular ROS levels (Fig. 9A and B). The therapeutic effects were antagonized by inhibitors of AMPK and SIRT1.

4. Discussion

AD poses significant challenges in treatment due to its complex neurological nature. Following extensive drug development, the

treatment of AD has evolved from a singular focus on targeting A β or Tau proteins to a multifaceted approach addressing various mechanisms, including neuroinflammation and cerebrovascular dysfunction [67–69]. The therapeutic options have also expanded from single small-molecule drugs to include antibodies and combination therapies [69]. Given the insidious onset and protracted course of AD, long-term medication is often necessary. Combination therapies derived from natural medicinal plants generally exhibit fewer side effects, making them advantageous for long-term use [70]. Our study demonstrated that BZBS alleviated

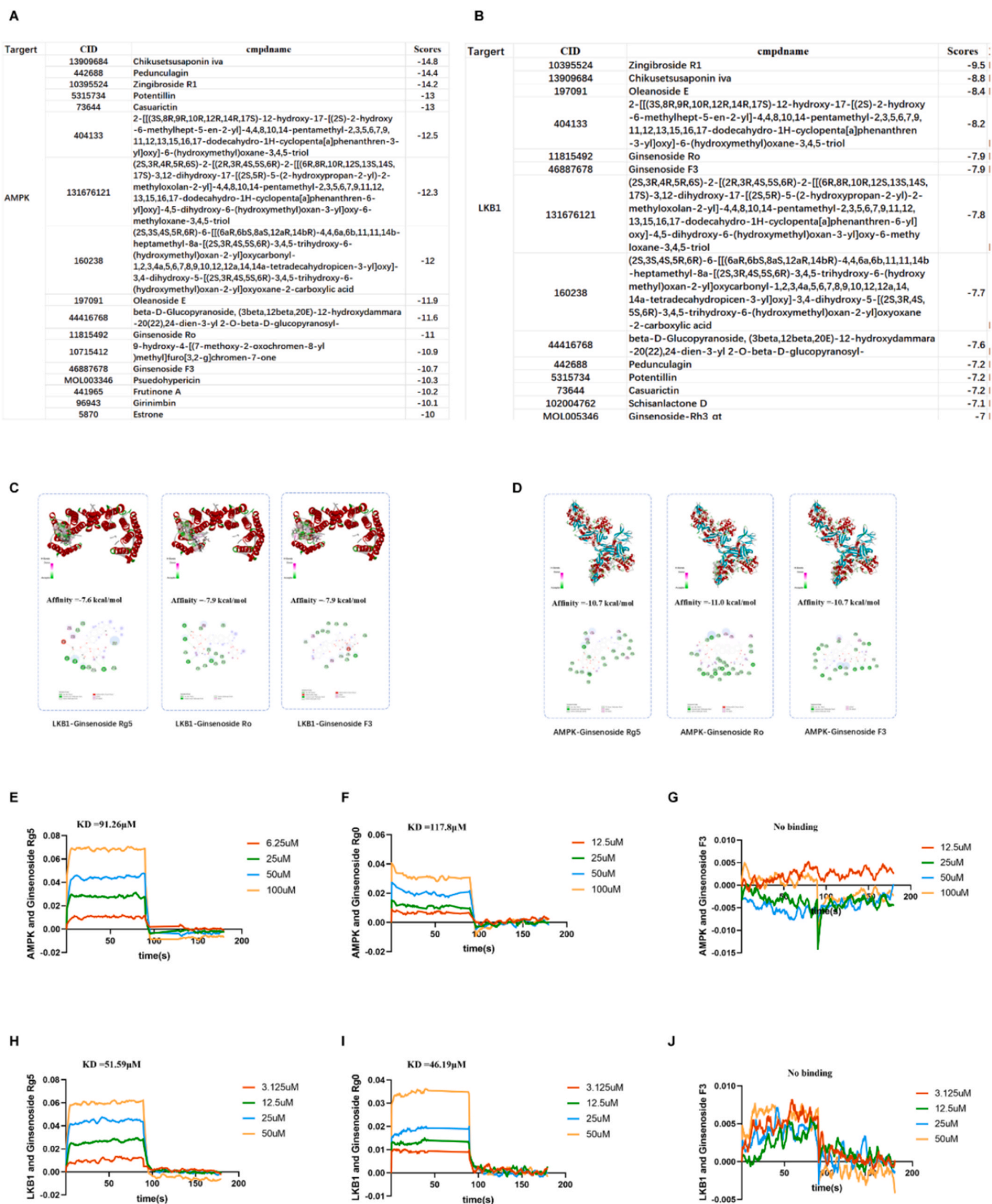


Fig. 7. Ginsenoside Rg5 and ginsenoside Ro target LKB1 and AMPK. (A–B) Compounds retrieved and scored by molecular docking of BZBS with LKB1 and AMPK. (C–D) Molecular docking of LKB1 and AMPK with ginsenoside Rg5, ginsenoside Ro, and ginsenoside F3. (E–G) BLI binding kinetics of the interaction between LKB1 and ginsenoside Rg5, ginsenoside Ro, and ginsenoside F3. (H–J) BLI binding kinetics of the interaction between AMPK and ginsenoside Rg5, ginsenoside Ro, and ginsenoside F3.

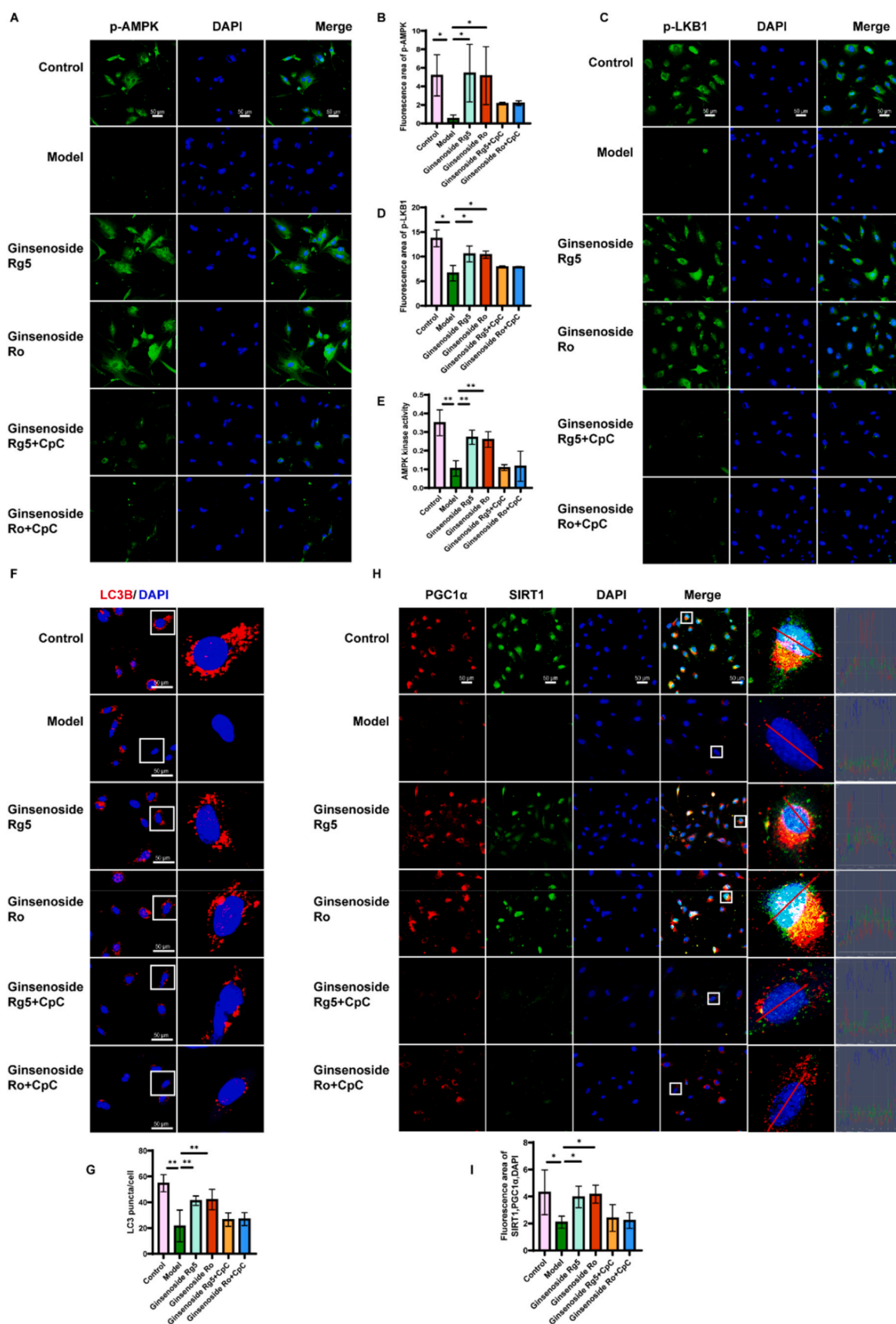


Fig. 8. Ginsenoside Rg5 and ginsenoside Ro enhance autophagy in bEnd.3 cells via the LKB1-AMPK-SIRT1 signaling pathway. (A–B) Expression of p-LKB1 in bEnd.3 cells was analyzed by immunofluorescence. (C–D) Expression of p-AMPK in bEnd.3 cells was analyzed by immunofluorescence. (E) Cellular AMPK kinase activity was measured in bEnd.3 cells. (F–G) LC3B positive vesicles in bEnd.3 cells were analyzed by immunofluorescence. (H–I) Co-expression of SIRT1, PGC1α, and DAPI in bEnd.3 cells was analyzed by immunofluorescence. $n = 3$; scale bar = 50 μm ; * $p < 0.05$, ** $p < 0.01$. Error bars represent SD.

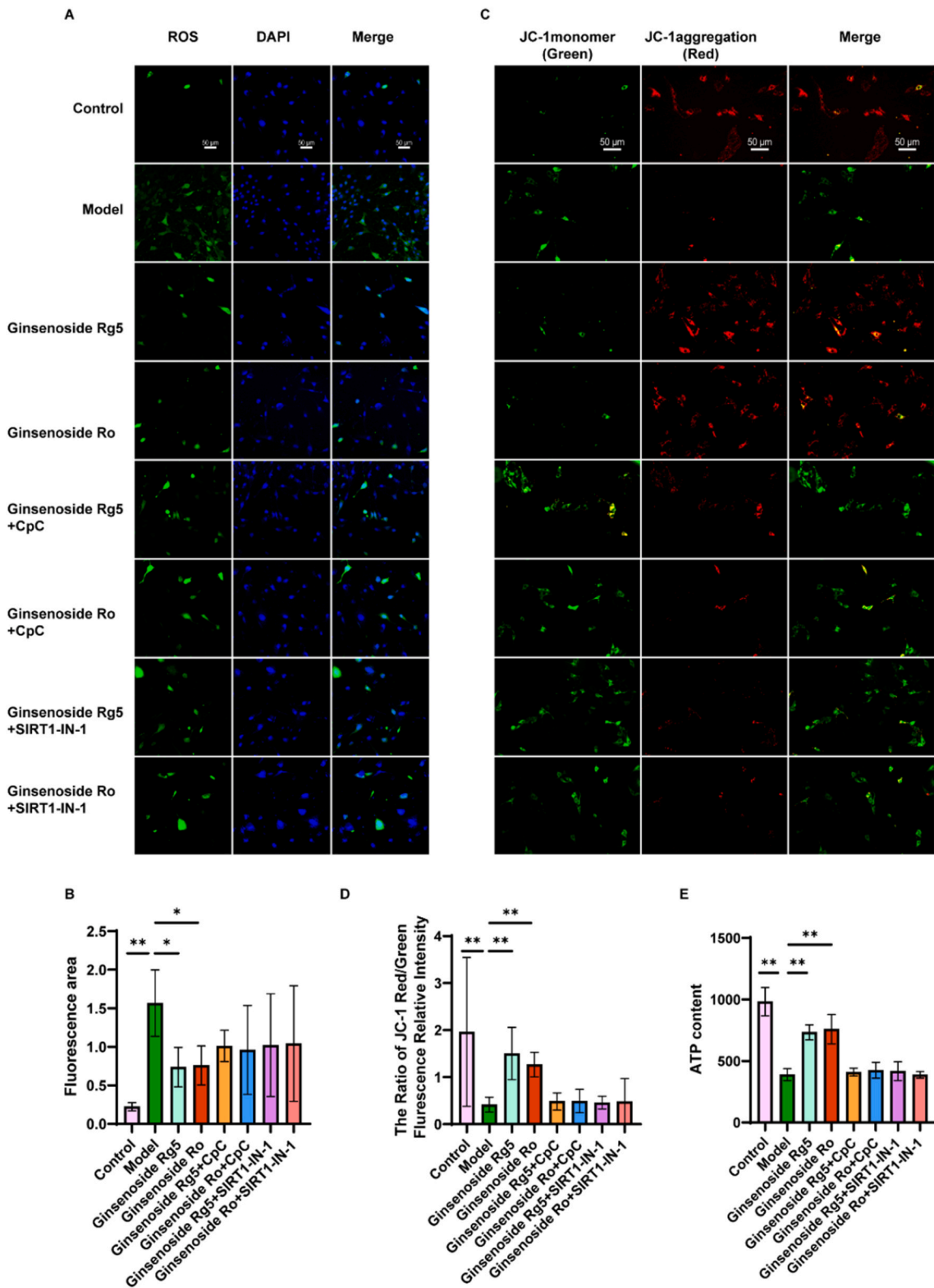


Fig. 9. Ginsenoside Rg5 and ginsenoside Ro improve mitochondrial dysfunction in bEnd.3 cells. (A–B) ROS assay in bEnd.3 cells. (C–D) Mitochondrial membrane potential assay in bEnd.3 cells. (E) ATP content assay in bEnd.3 cells. $n = 3$; scale bar = 50 μm ; * $p < 0.05$, ** $p < 0.01$. Error bars represent SD.

cognitive impairment in AD model mice by promoting cerebrovascular and mitochondrial autophagy, decreasing β -amyloid plaque deposition, enhancing mitochondrial function, and ultimately mitigating BBB damage and neuroinflammation. These findings further support the clinical application of natural compound botanicals in treating AD.

Donepezil has been shown to significantly enhance learning and

memory functions across various animal models [33,71,72]. Consequently, we chose donepezil as the positive control. The findings indicate that, regarding reducing BBB permeability and the activation of autophagy pathways, donepezil outperforms BZBS. However, BZBS surpasses donepezil in enhancing learning and memory, improving cerebral blood flow and cerebrovascular integrity, reducing A β deposition,

alleviating neuroinflammation, and decreasing oxidative stress levels (Fig. 1A–D, Fig. 1M–R, Fig. 2A–M and Fig. 3A–P). It is worth noting that BZBS is a compound medicine capable of simultaneously acting on multiple targets. Its therapeutic effects on Alzheimer's disease (AD) mice may involve multiple mechanisms. Further research is required to elucidate its comprehensive pharmacological mechanisms of action.

The importance of maintaining cerebrovascular endothelial homeostasis in AD and other neurodegenerative disorders has been overlooked. Recent research has indicated that maintaining cerebrovascular endothelial homeostasis is crucial for preserving brain health and influencing the progression of neurodegenerative disorders [73–75]. For example, Kiss et al. utilized single-cell sequencing technology to investigate the lineage expression of vascular endothelial cells in the mouse brain, and demonstrated substantial alterations in the number and functionality of specific subtypes as a result of aging [76]. Our findings further demonstrated that the BBB in AD mice was compromised, permitting substances such as serum albumin to infiltrate the brain and subsequently trigger neuroinflammation. This mechanism may be pivotal in the onset and progression of AD (Fig. 3I–K). Compared to $A\beta_{1-40}$, $A\beta_{1-42}$ plays a more significant role in the neurotoxicity associated with AD; however, $A\beta_{1-40}$ is the predominant isoform responsible for perivascular deposition and vascular damage. The accumulation of $A\beta_{1-40}$ elevates the levels of reactive ROS, which in turn causes lipid peroxidation, triggers cell apoptosis, and results in vessels damage [77]. Furthermore, $A\beta_{1-40}$ can activate MMPs, including MMP-2 and MMP-9, which degrade extracellular matrix proteins, tight junction proteins, and basement membrane proteins [78]. Consequently, $A\beta_{1-40}$ deposition is more closely associated with damage to the blood-brain barrier, attracting greater attention in scientific research. Furthermore, inadequate cerebral blood flow and dysfunction of the BBB represent early manifestations of neurodegenerative diseases in both humans and animal models [79–83]. Disruption of cerebrovascular endothelial homeostasis affects both brain blood flow and metabolism, leading to a reduced supply of oxygen and nutrients to the brain [83]. This situation increases the accumulation of harmful substances, which accelerates neuronal damage and cell death [83]. Recently, researchers have developed a series of drugs aimed at treating neurodegenerative diseases by targeting cerebrovascular endothelial homeostasis [84]. For example, clinical trials have been conducted on several drugs to treat AD by improving cerebral circulation [85,86]. These include telmisartan + perindopril (a small-molecule combination), angiotensin receptor blockers, and Yangxue Qingnao pills (a compound herbal medicine enhancing BBB integrity) [84]. A deeper understanding of cerebrovascular endothelial homeostasis in neurodegenerative diseases, alongside targeted therapies, may offer promising strategies for treating AD.

Metabolism and autophagy play crucial roles in maintaining cellular homeostasis [87]. Therapeutic effects associated with autophagy and mitochondrial autophagy have been documented in various diseases, including Parkinson disease, cerebral apoplexy, and Huntington disease [88–90]. In AD, increased levels of $A\beta$ and phosphorylated tau protein (p-tau) can induce ROS production, causing excessive fragmentation of mitochondria and promoting defective mitophagy [91]. Impaired mitochondrial dynamics (increased fission–reduced fusion, or vice versa) and defective mitophagy conditions in turn contribute to the synaptic dysfunction and cognitive deficits by triggering $A\beta$ and p-tau accumulation in the disease process of AD [91,92]. The LKB1–AMPK–SIRT1 connects energy status to various cellular functions, such as autophagy, lipid storage, and inflammatory response. Recent studies have demonstrated that AMPK activity decreases in AD brain, indicating decreased mitochondrial biogenesis and function [7]. Ginsenosides, main ingredients in BZBS, have multiple pharmacological

activities and less toxic side effects, and have shown beneficial effects in AD models *in vitro* and *in vivo* [90,93–95]. Ginsenoside Rg5 has been reported to regulate energy metabolism via AMPK-related signaling pathways [96]. However, the specific target is unknown. Limited research has been conducted on ginsenoside Ro. Our experiments demonstrated that ginsenosides Rg5 and Ro directly bind to LKB1 and activate the LKB1–AMPK–SIRT1 signaling pathway, promoting both cellular and mitochondrial autophagy (Fig. 7H–J, Fig. 8A–D and Fig. 9A–D). This synergistic effect represents a significant advantage of combination therapy. However, a comprehensive analysis of the pharmacological mechanisms of BZBS necessitates further exploration of its pharmacokinetics and pharmacodynamics.

5. Conclusion

Our study demonstrated that BZBS enhances autophagy-related activity, reduces $A\beta$ deposition, and improves endothelial cell homeostasis through the activation of the LKB1–AMPK–SIRT1 signaling pathway, which subsequently enhances cognitive function in AD mice. However, the dosage of BZBS employed in this study lacks comprehensive toxicological data, necessitating additional safety evaluations for its clinical application in treating AD. Nevertheless, this study lays the foundation for treating cognitive dysfunction in AD using combination therapies that target autophagy and endothelial cell homeostasis.

CRedit authorship contribution statement

Yawen Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Tongxing Wang:** Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis. **Hongrong Li:** Validation, Supervision, Software, Methodology, Conceptualization. **Yuning Jiang:** Visualization, Validation, Resources, Methodology. **Xiaogang Shen:** Software, Resources, Formal analysis, Data curation. **Ning Kang:** Methodology, Investigation, Formal analysis. **Zhifang Guo:** Supervision, Investigation, Formal analysis. **Runtao Zhang:** Methodology, Formal analysis. **Xuan Lu:** Software, Resources, Methodology. **Tianyu Kang:** Visualization. **Mengnan Li:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Funding acquisition, Conceptualization. **Yunlong Hou:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Funding acquisition. **Yiling Wu:** Writing – review & editing, Writing – original draft, Visualization, Data curation, Conceptualization.

Statement of professional ethics requirements

The Hebei Yiling Traditional Chinese Medicine Research Institute (Approval number: N2021174) gave its approval to all experimental methods.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.freeradbiomed.2025.03.045>.

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